Notes – 9 :- Bias, Varience etc

Intuition → part (claude)

- **Bias, varience, why generalization matters?**

⇒ This topic is about "you train a model on training data. Why should doing well on training data mean anything about new, unseen data"

- **Bias v/s Variance** :– (Imagine u trying to learn the shape of a river from 10 photos.)

- **High Bias (Underfitting)** :– U assume the river is always straight. Even with 1000 photos, u would be wrong – broken assumption. (Less precise – more accurate)

- **High Variance (overfitting)** :– You memorize every bend in those 10 photos – including the ones caused by temporary flooding. with new photos u'r pred. is wrong. (more precise, less accurate)

- **Key Insight** → • Bias = error from wrong assumptions → more data won't fix it

  • Variance = error from memorizing noise → more data will fix it.

• 2 tools collected →

1. **Union Bound** :→ P(atleast one of k bad events happens) ≤ sum of individual probabilities

   Ex → P(it rains or there is traffic) ≤ P(rain) + P(traffic)

2. **Hoeffding Inequality** :– If you flip a coin m times, your observed fraction of heads is close to the true probability – and the gap shrinks exponentially with m.

   Formula : $P(|\phi - \hat{\phi}| > \gamma) \leq 2 \cdot \exp(-2\gamma^2 m)$

   Concrete : flip 1000 times, $\gamma = 0.05 \rightarrow P(\text{error} > 5\%) \leq 2e^{(-5)} \approx 0.013$

: Note → your fault ← Bias stays always 3 b/z it comes from wrong model structure.

Varience goes down with more training data
data's fault → (more data = less noise memorization)

② Section-3: The Finite $\mathcal{H}$ case (heart of the notes).

⇒ Setup (Problem statement & assumptions):-
we have k hypothesis to choose from $\mathcal{H} = \{h_1, h_2, \ldots, h_k\}$
You pick the one with lowest training error

Two errors to distinguish:-

Symbol	Name	Meaning
$\hat{\varepsilon}(h)$	training	fraction wrong on your dataset
$\varepsilon(h)$	Generalization error / true error	Probability of wrong on new data

Fear →
you pick $\hat{h}$ = best hypothesis on training data.
But what if training error is misleadingly low?
maybe $\hat{h}$ just got lucky on your 50 training examples but is actually ~~lucky~~ terrible.
The notes want to prove: training error ≈ Generalization error $\forall h$.

Step-1: One hypothesis at a time ⇒
Each training example is a coin flip - did h misclassify it? (1) or not (0)
Training error = average of m such coin flips = $\hat{\phi}/\hat{\varepsilon}$
True error = true coin bias = ϕ/ε

Apply Hoeffding:-
$$P(|\varepsilon(h_i) - \hat{\varepsilon}(h_i)| > \gamma) \leq 2 \cdot \exp(-2\gamma^2 m)$$

So, for any single 'h' training error is close to real error with probability.

Step 2: But we have k hypothesis!
Problem:- even if each h individually is fine (say 1% chance of being misleading), with k hypo

hypothesis, the chance that at least one is misleading gets larger.

This is where Union Bound enters.

P(any one of K hypothesis is badly estimated)

$$\leq K \times 2\exp(-2\gamma^2 m) = 2K \cdot \exp(-2\gamma^2 m)$$

$\underbrace{2e^{(\dots)} + 2e^{(\dots)} + \dots}_{K \text{ times}}$

Flip it → probability that all hypotheses are well estimated

$$\geq 1 - 2K \cdot \exp(-2\gamma^2 m)$$

• This is called uniform convergence — training error tracks real error for all h at once.

St-8: So how bad can our chosen $\hat{h}$ be?

Define $h^* =$ ~~best~~ best possible hypothesis (lowest true error).

• The Notes prove:

$$\varepsilon(\hat{h}) \leq \varepsilon(h^*) + 2\gamma$$

bias, convenience

meaning: Our chosen model is at most 2γ worse than the best possible model in H. Thats guarantee.

• The Sample Complexity Formula:-

How many examples m do you need to guarantee this?

$$m \geq \frac{1}{2\gamma^2} \log \frac{2K}{\delta}$$

γ = error tolerance
δ = chance of failure
confidence = $1-\delta$

Crucially: m grows only $\log(K)$ with number of hypotheses.

Doubling K from 100 to 200? You barely need more data.

• Ex → Say K = 100 hypotheses, you want $\gamma = 0.05$ (5% error tolerance), $\delta = 0.05$ (95% confidence).

$$m \geq (1/0.005) \times \log(200/0.05) = 200 \times \log(4000)$$
$$\approx 200 \times 8.3 \approx 1660 \text{ examples.}$$

Section - 4: → Infinite H & V.C Dimension:

• The problem with infinite H:

Linear classifiers have parameters ∈ R — real numbers; infinite possible values → infinite hypotheses. our previous formula breaks down because $\log(\infty) = \infty$.

So, the old bound is useless. We need a new way to measure "complexity of H".

• The 64-bit Hack (just an analogy for intuition)

computers store real numbers in 64 bits. So, d parameters → 64 d bits → at most $2^{(64d)}$ distinct hypotheses.

plug into old formula: $m \geq O(d/\gamma^2 \cdot \log(\frac{1}{\delta}))$ order of

Conclusion: sample complexity (m) is linear in d (number of parameters).

But this is unsatisfying - why should the bit representation matter for learning theory?

• The real one: VC Dimension:-

We need a measure of complexity that's intrinsic to H, not its parameterization.

• Key concept: shattering

H shatters a set of points S if H can produce every possible labeling of those points.

• Concrete example with 3 points: - If I give u 3 points and $2^3 = 8$ possible ways to label them (each point + or -) can ur hypothesis class find a line that separates correctly for all 8 labelings?

For linear classifiers in 2D → yes for 3 points, not for 4 points.

(*) VC Dimension Definition:-

VC(H) = size of the largest set H can shatter

For 2D linear classifiers: VC = 3

(*) Why VC Dim better than parameter count?

→ Remember the notes showed same H can be written with n+1 parameters OR 2n+2 parameters. parameters count changed but the hypothesis class didn't

VC dim stays same regardless of parameters

It measures what H can actually do, not how it's written.

• The Big Theorem (Imp in learning theory)

→ with probability ≥ 1-δ, ∀ h∈H:

$$|\varepsilon(h) - \hat{\varepsilon}(h)| \leq O\left(\sqrt{\frac{d}{m}\log\frac{m}{d} + \frac{1}{m}\log\frac{1}{\delta}}\right)$$

where $d = VC(H)$ → parameter (indirectly)

Translation: gap b/w training & real error shrinks as m grows, as long as VC dimension is finite.

Final conclusion: m needed to learn well ∝ VC(H) ∝ no of parameters.

Intuition Summary →

concept	meaning
VC Dim	how "expressive" is your model
High VC	can shatter more points → more complex → needs more data
Low VC	Simpler model → less data needed but may have high bias.

Extra on few topics like Shattering:-

• Shattering – Core Idea:-
H shatters S means: no matter how I assign +/− labels to those points, H can always find a classifier that gets them all right.

Labeling clarification:-
Each point gets a binary label: + or −
3 points → each can be + or − independently → $2^3 = 8$ combinations

(+,+,+), (+,+,−), (+,−,+), (−,+,+),
(+,−,−), (−,+,−), (−,−,+), (−,−,−)

H must correctly classify all 8, not just 3.
• (~~Doubt about~~ linear classifier in 2D → yes for 3 points, no for 4)
→ (position matters here → that key subtlety)

VC Dim says: does there EXIST at least one arrangement of d points that H can shatter?
For 4 points – you can arrange them favorably, but there will always be at least one labeling a line can't separate (XOR-like pattern).
No arrangement of 4 points works for all 16 labelings with a linear classifier.

For 3 points – there exists an arrangement where all 8 labelings work.
(One summary → Shattering = H is powerful enough to memorize any pattern on those points)

(*) Why say VCd parameters if parameterization doesn't matter?
2D lin classifier → 3 param $(\theta_0, \theta_1, \theta_2)$
→ 3 VC Dim
• It is defined independent of parameterization.
• But empirically / practically for most hypothesis class
VC Dim ≈ Parameter Count.